

Compare function types



1 The table of values for function P and function Q are provided:

- Function P

x	-2	-1	0	1	2
y	9	6	3	0	-3

- Function Q

x	0	1	2	3	4
y	6	3	2	3	6

- Determine what type of functions Function P and Function Q are.
- Sketch the graph of the functions on the same coordinate plane.
- As $x \rightarrow \infty$, determine which function will have the greater value.
- Determine if each function is increasing or decreasing on the domain $x < 2$.

2 Consider the following pair of functions:

- Function P

$$y = -12 + \frac{x}{10}$$

- Function Q

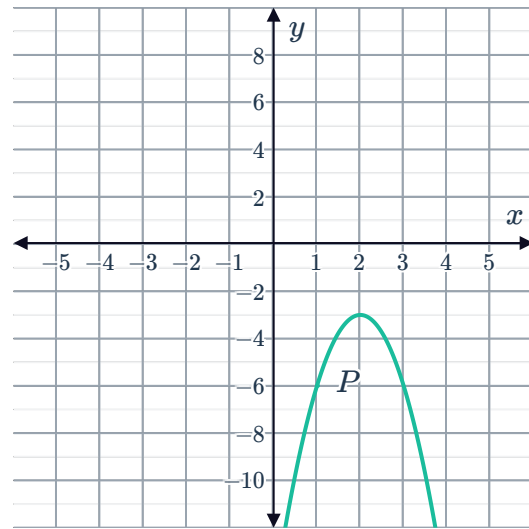
x	-8	-7	-6	-5	-4
y	-1	-4	-5	-4	-1

- Sketch the graph of the functions on the same coordinate plane.
- How many times do P and Q intersect?
- Determine which function has the greater function value at the y -intercept.
- Determine which function has the greater function value at $x = -1$.

3 The line P is given by $y = -4 + \frac{4x}{3}$ and the parabola Q is given by $y = -(x - 1)(x - 4)$.

- Sketch the graphs of line P and the parabola Q on the same coordinate plane.
- Identify how many times P and Q intersect.
- Identify which function has the greater function value at $x = 0$.

- 4 The graph of function P , given by the equation $y = -3(x - 2)^2 - 3$ is shown:



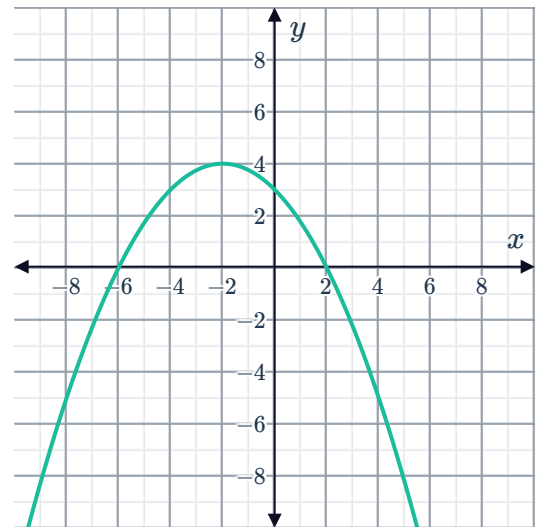
- Sketch the graph of function Q given by the equation $y = -6x + 12$ on the same coordinate plane.
- How many times do P and Q intersect?
- Determine which function has the greater function value at the y -intercept.
- Determine if each function is increasing or decreasing on the domain $x < 2$.

- 5 Determine which quadratic function has the greatest maximum value:

- Function P

x	-1	1	3
y	2	6	2

- Function Q



- 6 A table of values for parabola P and for parabola Q are provided:

- Parabola P

x	-8	-4	0
y	4	2	4

- Parabola Q

x	0	1	2
y	-5	-4	-5

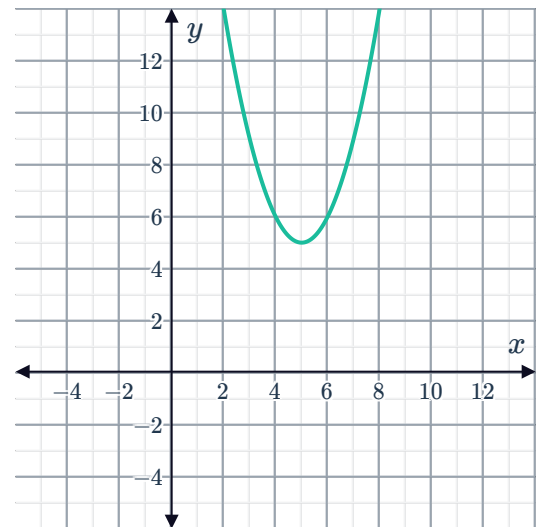
- Sketch the graph of parabolas P and Q on the same coordinate plane.
- Identify which parabola is concave up.
- Identify which parabola has the greater function value at the y -intercept.



7 Consider the following pair of functions:

- a Determine which function has a maximum of $y = 8$.
- b Determine which function has a minimum of $y = 5$.
- c Determine how many zeros function A has.
- d Determine how many zeros function B has.

- Function A
 $y = -2(x - 7)(x - 3)$
- Function B



8 Parabola P is given by $y = \frac{11(x - 6)^2}{36} - 6$ and parabola Q is given by $y = \frac{13(x - 3)^2}{9} - 8$.

- a Sketch the graph of parabolas P and Q on the same coordinate plane.
- b Determine which function has the lesser minimum.
- c Identify how many times the parabolas intersect.
- d Determine which parabola has the greater function value at $x = 8$.

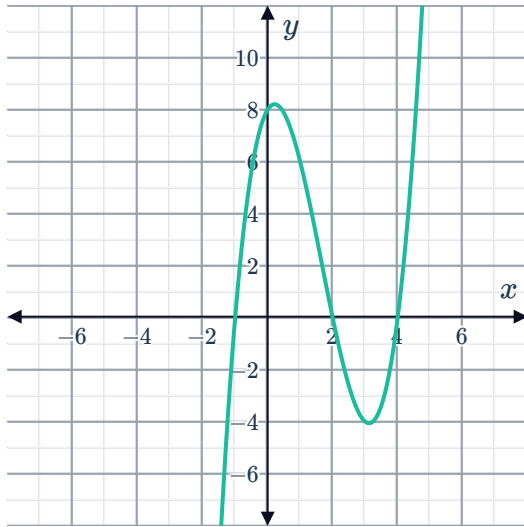
9 Cubic function P is given by $y = -(x + 2)^3 + 4$ and cubic function Q is given by $y = (x - 1)(x - 2)(x + 2)$.

- a Sketch the graph of cubic functions P and Q on the same coordinate plane.
- b Determine which cubic function has the greater function value at $x = -1$.
- c Determine how many zeros cubic function P has.
- d Determine how many zeros cubic function Q has.

10 Consider the following pair of cubic function



- Function P



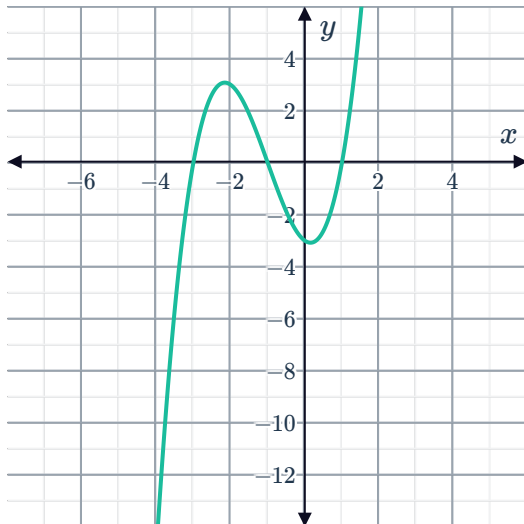
- Function Q

x	-5	-4	-3	-2
y	0	-6	-4	0

- Sketch the graph of function Q on the same coordinate plane as function P .
- Determine if each function is increasing or decreasing on the domain $1 < x < 3$.
- Describe the end behavior of each exponential function as $x \rightarrow \infty$.

11 Consider the following pair of cubic functions:

- Function P



- Function Q

$$y = -(x + 6)(x + 2)(x + 1)$$

- Sketch the graph of function Q on the same coordinate plane as function P .
- Determine which function has the greater function value at the y -intercept.
- Determine if each function is increasing or decreasing on the domain $-2 < x < -2$.
- Describe the end behavior of each exponential function as $x \rightarrow \infty$.



12 A table of values for the cubic function P and for the cubic function Q are provided:

- Cubic function P

x	-5	-4	-3	-2
y	0	6	4	0

- Cubic function Q

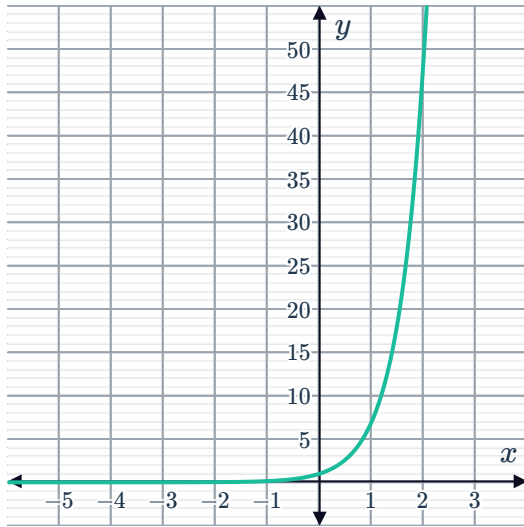
x	-1	0	1	2
y	0	9	8	3

- Sketch the graph of cubic functions P and Q on the same coordinate plane.
 - Determine which cubic function has the greater function value at the y -intercept.
 - Determine how many zeros cubic function P has.
 - Determine how many zeros cubic function Q has.
- 13 The exponential function J is given by $y = 9(3^{-x})$.
The following table shows the function values for exponential function K :
- | | | | | | |
|-----|----|---------------|---------------|-----------------|---------------------|
| x | -1 | 1 | 2 | 5 | 10 |
| y | 15 | $\frac{5}{3}$ | $\frac{5}{9}$ | $\frac{5}{243}$ | $\frac{5}{59\,049}$ |
- Determine if the exponential function K is increasing or decreasing.
 - Sketch the graph of exponential functions J and K on the same coordinate plane.
 - Describe the end behavior of each function as $x \rightarrow \infty$.
 - Determine what transformation we can apply to the function $y = 3^{-x}$ to produce function K .
- 14 The exponential function E is given by $y = 3^x + 4$ and the exponential function F is given by $y = -8^x + 1$.
- Sketch the graph of exponential functions E and F on the same coordinate plane.
 - Identify how many times E and F intersect.
 - Describe the end behavior of each exponential function as $x \rightarrow \infty$.

15 Consider the following pair of functions:



- Function P



- Function Q
 $y = 2^x + 4$

- Determine how many times the graphs of P and Q intersect.
- Determine which function tends to infinity the quickest as $x \rightarrow \infty$.

16 Consider the following pair of functions:

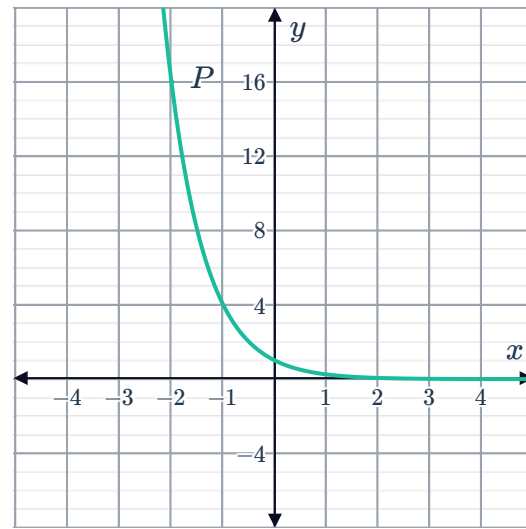
- Function P
 $y = 6^{-x} + 5$

- Function Q

x	-4	-3	-2	-1	0
y	0	$-\frac{11}{4}$	$-\frac{11}{2}$	$-\frac{33}{4}$	-11

- Determine what type of functions Function P and Function Q are.
- Sketch the graph of the functions on the same coordinate plane.
- Identify which function has the greater function value at the y -intercept.
- Identify how many times P and Q intersect.

- 17 The graph of the exponential function P , given by $y = 4^{-x}$ is shown on the right:



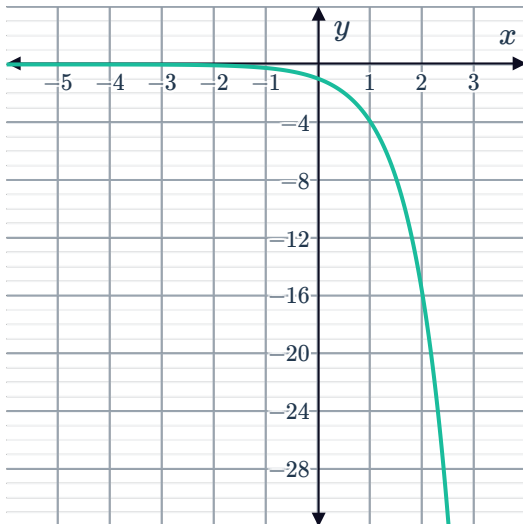
- Sketch the graph of function Q given by the equation $y = 8x + 12$ on the same coordinate plane.
- How many times do P and Q intersect?
- Which function has the greater function value at the y -intercept?
- Which function has the greater function value at $x = 1$?

- 18 The exponential function P is given by the equation $y = 5^{-x} + 3$ and the linear function Q is given by the equation $y = -4x + 4$.

- Sketch the graph of the functions on the same coordinate plane.
- How many times do P and Q intersect?
- As x approaches ∞ , which function is greater?

- 19 Consider the following pair of functions:

- Function A



- Function B

$$y = -\frac{4(x-3)(x-7)}{21}$$

- Determine which function has the greater function value at the y -intercept.
- Determine if each function is increasing or decreasing on the domain $0 < x < 5$.

20 The exponential function P is given by $y = 4^x$ and the parabola Q is given by

$$y = \frac{(x-6)(x-2)}{4}.$$

- Sketch the graph of exponential function P and the parabola Q on the same coordinate plane.
- Determine which function has the greater function value at $x = 4$.
- Determine if each function is increasing or decreasing on the domain $1 < x < 3$.

21 The table of values for the functions P and Q are provided:

- Function P

x	-7	-6	-5	-4	-3	-2
y	-7	-4	-3	-4	-7	-12

- Function Q

x	-10	-5	-1	1	5	10
y	1 073 741 824	32 768	8	$\frac{1}{8}$	$\frac{1}{32\,768}$	$\frac{1}{1\,073\,741\,824}$

- Determine what type of functions Function P and Function Q are.
- Sketch the graph of the functions on the same coordinate plane.
- Determine how many times functions P and Q intersect.



Applications

22 A table of values for the function P and a table of approximate values for the function Q is provided below:

- Function P

x	-4	-3	-2	-1	0	1	2
y	-3	0	3	6	9	12	15

- Function Q

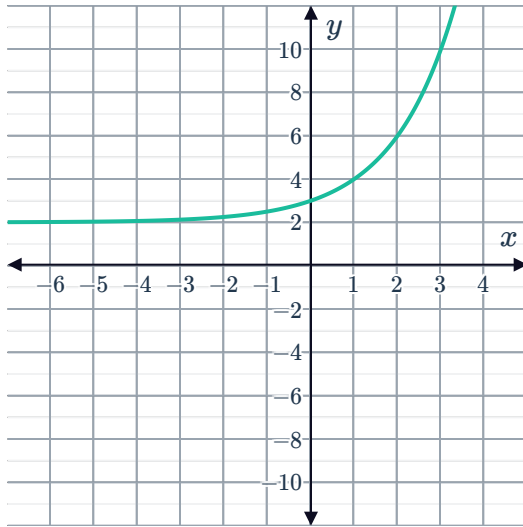
x	-5	-2	-1	0	1	5	10
y	-32 765	-61	-5	2	$2\frac{7}{8}$	$2\frac{32\,767}{32\,768}$	$2\frac{1\,073\,741\,823}{1\,073\,741\,824}$

- Which of the functions are linear and non-linear?
- Sketch the graph of function P .
- Given that Q is an exponential function, sketch the graph of the function Q on the same coordinate plane.
- Using part (c), how many solutions are there to the equation $P - Q = 0$?

23 Consider the graph of the exponential function  and the table of values for parabola Q :

• Function $P : y = 2^x + 2$

• Parabola $Q : y = -3(x + 5)(x + 2) + 2$



x	-5	-4	-3	-2	-1
y	2	8	8	2	-10

- Sketch the graphs of P and Q on the same coordinate plane.
- Describe the behavior of function P and function Q as x approaches infinity.
- Using the graph, determine how many solutions there are to the equation $2^x + 2 + 3(x + 5)(x + 2) - 2 = 0$.

24 The population of two different bacteria, labeled J and K , are given by the following table of values:

Bacteria J :

t (time in days)	0	1	2	3	4
P (population)	1	60	3600	2.16×10^5	1.296×10^7

Bacteria K :

t (time in days)	0	1	2	3	4
Q (population)	1	50	2500	1.25×10^5	6.25×10^6

- Determine which population of bacteria increases faster.
- The population P of bacteria J at time t can be modeled using the standard equation $P(t) = a^t$, where $a > 1$. By using the table of values, graph $P(t)$ for $t > 0$.
- The population Q of bacteria K at time t can be modeled using the standard equation $Q(t) = b^t$, where $b > 1$. By using the table of values, graph $Q(t)$ for $t > 0$.